```
!mkdir FUTURE_ML_01
```

```
!ls
```

```
FUTURE_ML_01  sample_data
```

```
!mkdir FUTURE_ML_01/images
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
```

```
df = pd.read_csv("/content/FUTURE_ML_01/walmart/Walmart.csv")

df.head()
```

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
0	1	05-02-2010	1643690.90	0	42.31	2.572	211.096358	8.106
1	1	12-02-2010	1641957.44	1	38.51	2.548	211.242170	8.106
2	1	19-02-2010	1611968.17	0	39.93	2.514	211.289143	8.106
3	1	26-02-2010	1409727.59	0	46.63	2.561	211.319643	8.106
4	1	05-03-2010	1554806.68	0	46.50	2.625	211.350143	8.106

```
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

df.dropna(inplace=True)

print("\nAfter Cleaning:")
print(df.shape)
```

```
(6435, 8)

Missing Values:
Store           0
Date            0
Weekly_Sales    0
Holiday_Flag    0
Temperature     0
Fuel_Price      0
CPI             0
Unemployment    0
dtype: int64

After Cleaning:
(6435, 8)
```

```
df['Date'] = pd.to_datetime(
    df['Date'],
    format='%d-%m-%Y'
)

df.head()
```

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
0	1	2010-02-05	1643690.90	0	42.31	2.572	211.096358	8.106
1	1	2010-02-12	1641957.44	1	38.51	2.548	211.242170	8.106
2	1	2010-02-19	1611968.17	0	39.93	2.514	211.289143	8.106
3	1	2010-02-26	1409727.59	0	46.63	2.561	211.319643	8.106
4	1	2010-03-05	1554806.68	0	46.50	2.625	211.350143	8.106

```
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['Day'] = df['Date'].dt.day

df.head()
```

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment	Year	Month	Day
0	1	2010-02-05	1643690.90	0	42.31	2.572	211.096358	8.106	2010	2	5
1	1	2010-02-12	1641957.44	1	38.51	2.548	211.242170	8.106	2010	2	12
2	1	2010-02-19	1611968.17	0	39.93	2.514	211.289143	8.106	2010	2	19
3	1	2010-02-26	1409727.59	0	46.63	2.561	211.319643	8.106	2010	2	26
4	1	2010-03-05	1554806.68	0	46.50	2.625	211.350143	8.106	2010	3	5

```
X = df[
    [
        'Store',
        'Holiday_Flag',
        'Temperature',
        'Fuel_Price',
        'CPI',
        'Unemployment',
        'Year',
        'Month',
        'Day'
    ]
]

y = df['Weekly_Sales']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)
```

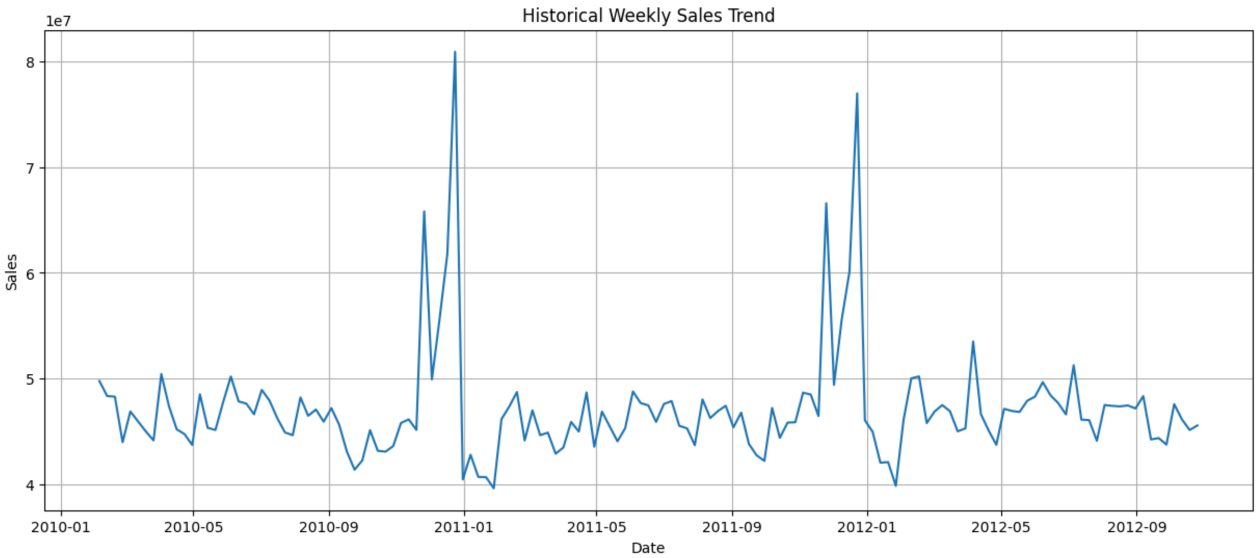
```
▾     RandomForestRegressor     i ?
RandomForestRegressor(random_state=42)
```

```
pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, pred)

mse = mean_squared_error(y_test, pred)

rmse = np.sqrt(mse)

r2 = r2_score(y_test, pred)

print("MAE :", mae)

print("RMSE :", rmse)

print("R2 Score :", r2)
```

```
MAE : 58078.98642338773
RMSE : 116210.05824650699
R2 Score : 0.958079819067563
```

```
plt.figure(figsize=(15,6))

sales_by_date = df.groupby('Date')['Weekly_Sales'].sum()

plt.plot(
    sales_by_date.index,
    sales_by_date.values
)

plt.title("Historical Weekly Sales Trend")

plt.xlabel("Date")

plt.ylabel("Sales")

plt.grid()

plt.show()
```



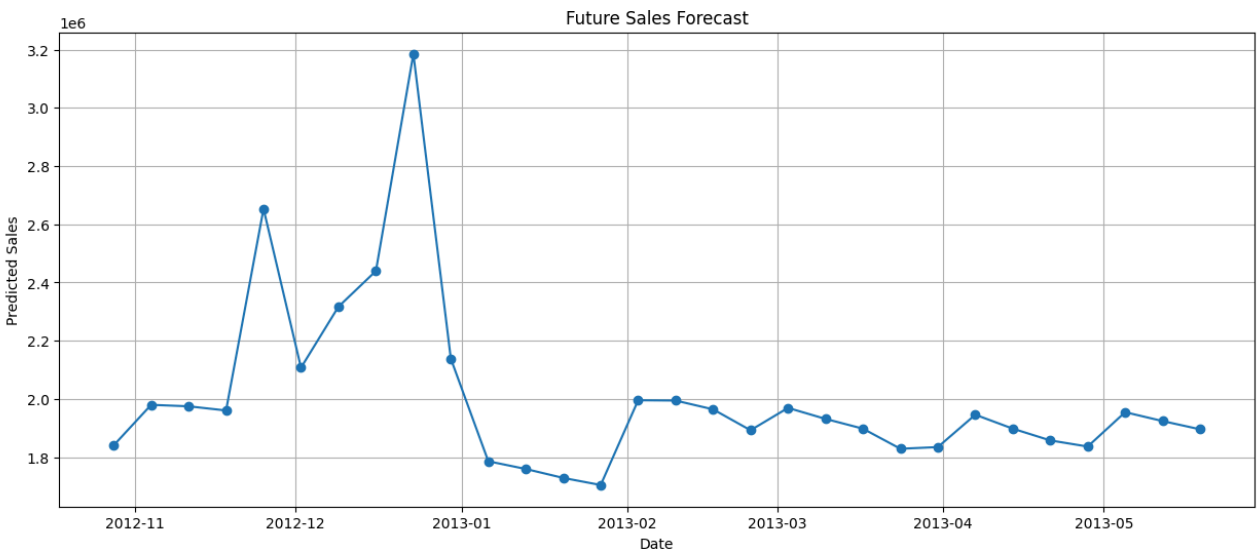

```
future_dates = pd.date_range(
    start=df['Date'].max(),
    periods=30,
    freq='W'
)

future_df = pd.DataFrame()

future_df['Date'] = future_dates

future_df['Store'] = 1

future_df['Holiday_Flag'] = 0

future_df['Temperature'] = df['Temperature'].mean()

future_df['Fuel_Price'] = df['Fuel_Price'].mean()

future_df['CPI'] = df['CPI'].mean()

future_df['Unemployment'] = df['Unemployment'].mean()

future_df['Year'] = future_df['Date'].dt.year

future_df['Month'] = future_df['Date'].dt.month

future_df['Day'] = future_df['Date'].dt.day
```

```
future_sales = model.predict(
    future_df[
        [
            'Store',
```

```
            'Holiday_Flag',
            'Temperature',
            'Fuel_Price',
            'CPI',
            'Unemployment',
            'Year',
            'Month',
            'Day'
        ]
    ]
)

future_df['Forecasted_Sales'] = future_sales

future_df.head()
```

	Date	Store	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment	Year	Month	Day	Forecasted_Sales
0	2012-10-28	1	0	60.663782	3.358607	171.578394	7.999151	2012	10	28	1.841104e+06
1	2012-11-04	1	0	60.663782	3.358607	171.578394	7.999151	2012	11	4	1.980154e+06
2	2012-11-11	1	0	60.663782	3.358607	171.578394	7.999151	2012	11	11	1.975112e+06
3	2012-11-18	1	0	60.663782	3.358607	171.578394	7.999151	2012	11	18	1.960789e+06
4	2012-11-25	1	0	60.663782	3.358607	171.578394	7.999151	2012	11	25	2.653232e+06

```
plt.figure(figsize=(15,6))

plt.plot(
    future_df['Date'],
    future_df['Forecasted_Sales'],
    marker='o'
)

plt.title("Future Sales Forecast")

plt.xlabel("Date")

plt.ylabel("Predicted Sales")

plt.grid()

plt.show()
```




```
readme_content = """
# Sales & Demand Forecasting for Businesses

## Objective
Build a Machine Learning model to forecast future sales using historical Walmart sales data.

## Dataset
Walmart Weekly Sales Dataset

## Tools Used
- Python
```

```
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

## Project Workflow
1. Data Collection
2. Data Cleaning
3. Feature Engineering
4. Exploratory Data Analysis
5. Model Building using Random Forest Regressor
6. Model Evaluation
7. Future Sales Forecasting
8. Visualization of Results

## Features Used
- Store
- Holiday_Flag
- Temperature
- Fuel_Price
- CPI
- Unemployment
- Year
- Month
- Day

## Evaluation Metrics
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R² Score

## Business Insights
- Sales trends vary across different stores.
- Holiday periods significantly affect sales.
- Economic indicators such as CPI and unemployment impact demand.
- Forecasting helps optimize inventory and business planning.
- Data-driven decisions can improve operational efficiency.

## Results
The Random Forest model was trained on historical Walmart sales data and used to forecast future sales trends. The model pe

## Author
Tejaswini Malladi
Future Interns - Machine Learning Internship
"""

with open("README.md", "w") as file:
    file.write(readme_content)

print("README.md created successfully!")
```

```
README.md created successfully!
```

```
!ls
```

```
FUTURE_ML_01  README.md  sample_data
```

```
from google.colab import files
files.download("README.md")
```

```
from google.colab import files
files.download("/content/FUTURE_ML_01/walmart/Walmart.csv")
```

```
plt.savefig("sales_trend.png")
plt.savefig("forecast.png")
```

```
<Figure size 640x480 with 0 Axes>
```

```
!ls
```

```
forecast.png  FUTURE_ML_01  README.md  sales_trend.png  sample_data
```